

Notes:
J. End stop sensors: Pin 1 is GND, 2 is Signal and 3 is +5VDC.

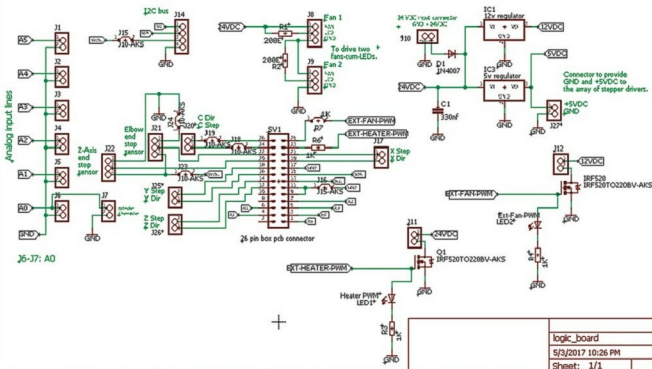


Fig. 2: The full schematic

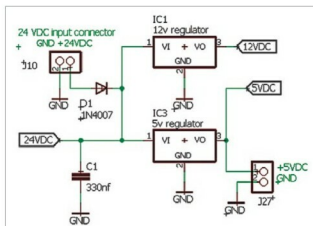


Fig. 3: The schematic zoomed-in view of the upper right side of the full schematic

If the components are handy, you may even use a general-purpose breadboard to quickly wire the circuit and test it. So let's say that you have the circuit for which we wish to create the schematic. At this stage, you may want to consider

using some kind of software tool for drawing the schematic on the computer. As you will soon realise, there are several advantages to using this tool as it works well beyond merely as a drafting or drawing aid. Eagle CAD, in particular, is extremely useful as it helps you create great schematics and much more. Its free version has some limitations like the maximum PCB size (board size) of 10x8cm² and only two signal layers allowed. Even with these limitations, the tool is sufficient for most constructional projects.

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Eagle comes with loads of goodies like a huge device library, several third-party libraries from manufacturers, built-in scripts that can perform many critical static checks, generate bill-of-materials and Gerber files that can be sent to a PCB manufacturing house, and much more. You can find loads of information and tutorials on YouTube and other web resources about Eagle CAD and similar software tools. Note that Eagle tool is used here for illustration purpose only, and you may use any other suitable tool of your liking, like Proteus.

By now you should have a schematic like the one in Fig. 2. Here Fig. 3 is a close-up of the top right corner of Fig. 2. Eagle also provides a very useful feature called 'electrical rule check' (ERC), which can be used to test